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Fundamental and practical efficiency limits of selenium solar cells under outdoor and indoor illumination

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ABSTRACT

Selenium (Se), particularly its trigonal phase, offers a wide-bandgap platform for outdoor and indoor photovoltaics. This work quantifies its spectrum-dependent efficiency limits using reciprocal optical absorption, radiative and nonradiative recombination, finite thickness, and electrical resistance, with separate module-temperature and degradation sensitivities. At a bandgap of 1.85 eV and 300 K, the one-sun air mass 1.5 global Shockley–Queisser limit is 26.10%; the nominal 500-lux limits are 52.86% for a white light-emitting diode (LED) and 53.16% for a compact fluorescent lamp. A representative experimental absorption spectrum gives one-micrometer-thick planar and textured radiative limits of 26.71% and 28.00%, respectively. A non-unique calibration reproduces three independent performance metrics of the certified 10.3% device. Parameter-conditioned advanced-target and practical-limit scenarios give 20.5%–26.9% outdoors and 26.5%–42.2% under the LED spectrum. These are isothermal, external-response-conditioned results, not forecasts of device performance. The full Shockley–Read–Hall expression and finite-doping calculations distinguish low-injection behavior from the intrinsic baseline. The radiative and recombination maps assume spatially uniform carrier densities and quasi-Fermi-level splitting through the absorber, together with complete extraction unless an empirical external-response factor is stated; they are zero-dimensional optical-recombination limits rather than drift–diffusion predictions for low-mobility Se films. Temperature-dependent optical-edge shifts are treated as illustrative sensitivities rather than calibrated material parameters. An ideal sub-bandgap reflector can reject 56.3% of incident solar power if an opaque module would otherwise absorb it, compared with only 1.30% under the modeled LED. The analysis identifies distinct outdoor and indoor priorities in absorption-edge quality, recombination, carrier extraction, leakage, and rear-stack optics, and provides reproducible parameter maps for interpreting future Se measurements.

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I. INTRODUCTION

The detailed-balance limit introduced by Shockley and Queisser (SQ) remains the canonical thermodynamic benchmark for a single-junction photovoltaic absorber because it ties absorption, spontaneous emission, electrical work, and entropy generation to the same optical transition probability.[1–3] Its utility lies not only in the familiar peak efficiency near a bandgap of 1.34 eV, but also in the way the framework can be generalized to real absorptivity, restricted optical *etendue*, nonradiative recombination, luminescence extraction, resistive transport, and non-solar spectra.[4–8] Extensions beyond reciprocal single-junction operation include finite-stage nonreciprocal radiative routing toward Landsberg-type benchmarks and carrier-multiplication/hot-carrier limits with explicit accounting of the shared above-gap energy reservoir.[9,10] These extensions are especially important for materials whose useful application is determined by spectral context rather than by the one-sun optimum.

Trigonal selenium (t-Se) is such a material. Se enabled the earliest solid-state photovoltaic experiments and the first thin-film solar cell, but its approximately 1.8–2.0 eV optical-gap range restricted the photocurrent available under terrestrial sunlight.

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Modern processing revived the platform: careful selection of electron- and hole-selective contacts, ultrathin tellurium nucleation layers, controlled crystallization, closed-space annealing, and illumination-assisted annealing have increased certified power-conversion efficiency from the historical 5% plateau to 10.3%, with $V_{OC} > 1$ V.[11–14] The current record nevertheless retains a voltage deficit exceeding approximately 0.55 V, an external quantum efficiency (EQE) onset substantially broader than the photoluminescence edge, and a short-circuit current well below the absorptivity-limited value.[14] The result is a material with an attractive thermodynamic limit, although its dominant loss channels remain incompletely separated. Related material-constrained multijunction calculations for transfer-printed van der Waals absorbers show that accessible bandgap windows, external radiative quality, luminescent coupling, and optical boundary conditions determine how rapidly the efficiency gain saturates toward the multicolor limit.[15]

The same wide bandgap that penalizes one-sun current is close to optimal for indoor photovoltaics. White light-emitting diodes (LEDs) and compact fluorescent lamps (CFLs) are designed around human photopic vision and concentrate most radiative power within a wavelength range of approximately 400–700 nm. Unlike sunlight, they contain little near-infrared power that lies below the t-Se gap. Se devices have consequently reached 15.1% under a 2700 K LED at 1000 lux and maintained output for 1000 h in ambient conditions.[16] Spectrum-resolved detailed-balance calculations for indoor energy harvesting further show that the optimum bandgap shifts toward approximately 1.8–2.0 eV, and that efficiency cannot be compared at a stated illuminance unless the source spectrum and irradiance are both specified.[17–19]

A second, less explored consequence of the large t-Se bandgap is thermal. For an opaque module that parasitically absorbs transmitted photons, more than half of air mass 1.5 global (AM1.5G) incident power lies below the t-Se gap. A spectrally selective sub-bandgap reflector (SBR) could reject this unusable power, reduce module temperature, improve voltage, and decelerate thermally activated degradation. The recent optical–temperature–electrical treatment of SBR-equipped modules demonstrates that such benefits must be evaluated together with above-bandgap transmission loss, mounting geometry, wind speed, ambient temperature, and energy yield rather than by optical reflectance alone.[20,21] The situation is qualitatively different indoors: LED and CFL irradiances at hundreds of lux are only a few watts per square meter, so self-heating and near-infrared absorption are intrinsically small.

Here, I establish a Se-specific framework for the fundamental and practical efficiency limits of single-junction t-Se photovoltaics under outdoor and indoor illumination. The framework resolves how the experimentally derived Se absorption edge, finite-thickness optics, carrier-recombination quality, source spectrum, and electrical parasitics determine distinct outdoor and indoor limits, and evaluates their implications for thermal management through separately stated module-level sensitivities. The model assumes spatially uniform carrier densities and quasi-Fermi-level splitting throughout the absorber, so volumetric Shockley–Read–Hall (SRH) and Auger rates are integrated as qdU, and photogenerated current is completely extracted unless a separately defined external-response factor is applied. This zero-dimensional approximation does not solve Poisson or drift–diffusion equations, nonuniform generation and fields, depletion-width evolution, diffusion lengths, or contact boundary conditions. Consequently, the calculated maps are optical–recombination upper limits and parameter sensitivities, not predictions of transport-limited performance in low-mobility Se devices.[14,22] The framework is anchored to the certified Se record and converts the governing optical, electrical, and thermal coordinates into experimentally testable parameter maps, thereby going beyond a bandgap-only benchmark. The main objectives are to: (i) determine spectrum-specific SQ limits and the variables that shift them; (ii) replace the step absorber with a representative experimental absorption coefficient and finite-thickness optics; (iii) quantify radiative, SRH, Auger, photon-escape, mirror, and resistive losses under the stated uniform-carrier approximation; (iv) calculate conditional outdoor and indoor scenarios anchored to the certified record; and (v) connect sub-bandgap rejection to module temperature and illustrative degradation kinetics. The explicit assumptions and derivations are given in Sec. S1 of the supplementary material; spectral and photometric normalization in Sec. S2 of the supplementary material; numerical convergence in Sec. S3 of the supplementary material; and the optical-data digitization and uncertainty protocol in Sec. S4 of the supplementary material.

II. RESULTS

A. A unified spectrum-to-yield framework for t-Se

Figure 1 defines the analysis space. Figure 1(a) compares one-sun AM1.5G, a white LED, a CFL, and a halogen lamp after transforming the wavelength-domain irradiance I_λ to the photon-energy-domain density $I_E = I_\lambda |d\lambda/dE|$ and plotting the normalized density $\tilde{I}_E(E) = I_E(E)/\int I_E(E')\,dE'$ on a semilogarithmic ordinate. The quotient has units eV^{-1} and is used only to compare a spectral shape. The shaded 1.80–1.90 eV window marks the experimentally relevant t-Se band-edge range. Sunlight spans both high-energy photons that thermalize and a broad sub-bandgap near-infrared reservoir, whereas the LED and CFL concentrate power close to and above the t-Se edge. Figure 1(b) shows the same indoor spectra on a wavelength axis at 500 lux and reports their very different irradiances: approximately 1.75 W m^{-2} for the LED, 1.45 W m^{-2} for the CFL, and 26.0 W m^{-2} for the halogen source. The dashed line at $\lambda_g = hc/E_g$, corresponding to a wavelength of approximately 670 nm, separates photons that can and cannot be absorbed by a 1.85 eV ideal absorber.

Figure 1(c) introduces the material optical model. A representative experimental t-Se absorption coefficient, digitized from the compilation in the 2026 review,[14,23] rises from approximately 10^3 to above 10^5 cm^{-1} across the edge. The corresponding two-pass absorptivity depends strongly on thickness near the band edge even though high-energy photons are absorbed in tens of nanometers. Figure 1(d) summarizes the sequential calculation: the illumination spectrum sets generation; the optical stack sets reciprocal absorptivity and emissivity; the electrical model sets the current–voltage curve; module temperature feeds back into emission and voltage; and temperature-dependent degradation maps instantaneous power onto lifetime energy yield. The diagram denotes a

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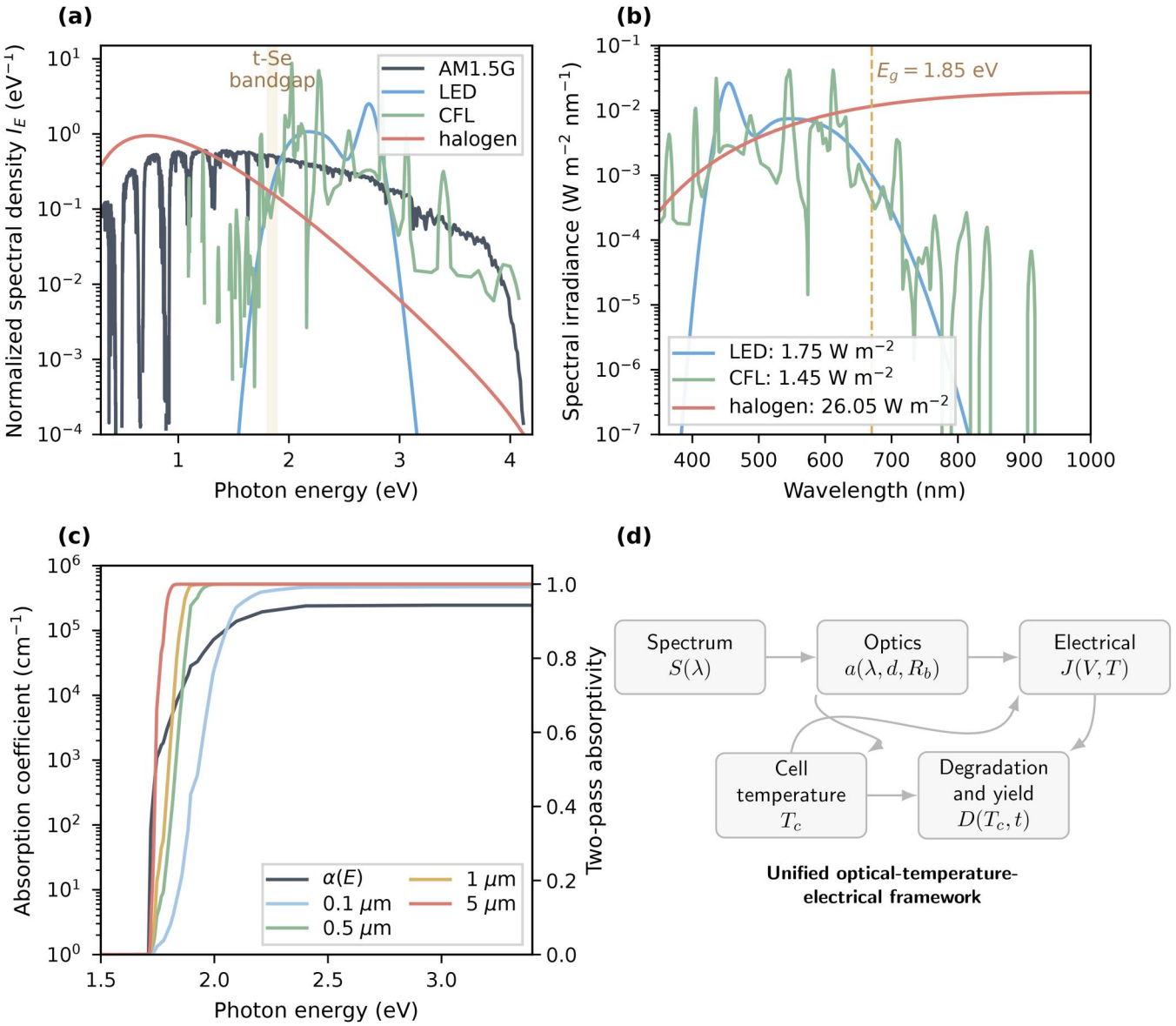


FIG. 1. Unified optical–temperature–electrical framework for Se photovoltaics. (a) Semilogarithmic comparison of the normalized photon-energy-domain spectral density $\tilde{I}_E(E) = I_E(E)/\int I_E(E')\,dE'$, where $I_E = I_\lambda|d\lambda/dE|$, for AM1.5G sunlight, a white light-emitting diode (LED), a compact fluorescent lamp (CFL), and a halogen lamp. The normalization compares spectral shape rather than absolute irradiance. The shaded band marks 1.80–1.90 eV, the principal optical-gap range reported for trigonal Se (t-Se). (b) Absolute spectral irradiance of the three indoor sources at 500 lux. The vertical dashed line is the wavelength corresponding to $E_g = 1.85$ eV. The source-dependent irradiance demonstrates why lux alone is not an energetic normalization. (c) Representative experimental t-Se absorption coefficient (left axis) and calculated two-pass absorptivity (right axis) for absorber thicknesses from 0.1 to 5 μ m. Near-edge absorption, rather than visible-region absorption, controls the thickness required for current saturation. (d) Calculation workflow coupling spectrum, reciprocal optics, detailed-balance electrical response, module temperature, and thermally accelerated degradation. The arrows indicate a sequential calculation: optical inputs determine generation and the effective thermal load, temperature enters the electrical calculation, and electrical output and temperature enter an illustrative yield estimate. No self-consistent thermal transport or site-resolved annual-yield simulation is claimed.

sequence of conditional models rather than a self-consistent energy-balance loop. The normalized remaining performance is $D(T_c, t) = \exp[-r(T_c)t]$, used only for the illustrative aging calculation.

B. Spectrum-specific SQ limits

Figure 2(a) gives the ideal step-absorber efficiency as a function of bandgap. The AM1.5G maximum is 33.68% at

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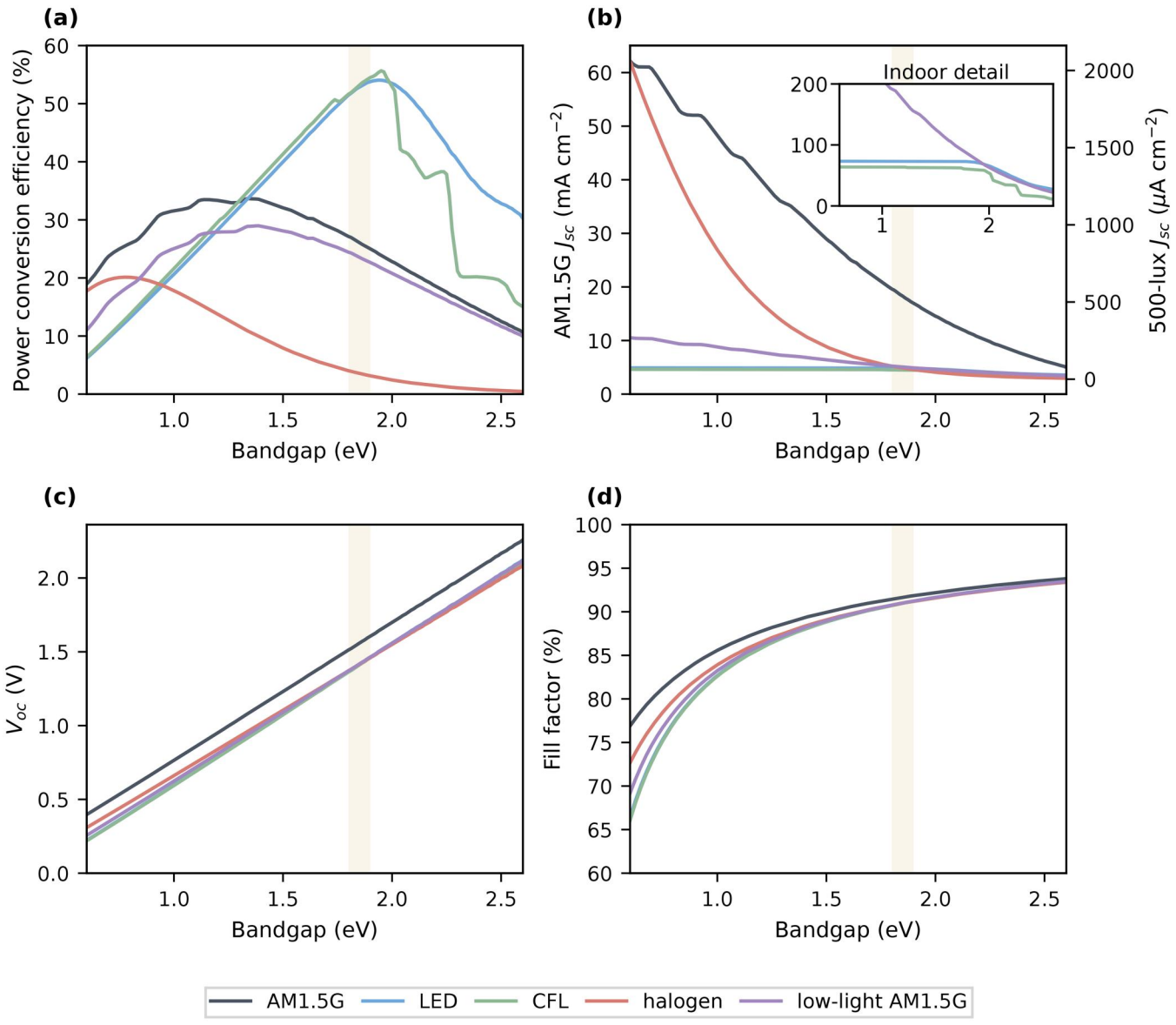


FIG. 2. Spectrum-dependent ideal detailed-balance limits. (a) Power-conversion efficiency vs bandgap under one-sun AM1.5G, LED, CFL, halogen, and a low-light AM1.5G spectrum normalized to 500 lux. Indoor LED and CFL optima occur near the t-Se gap and exceed 50%, whereas halogen illumination is dominated by unusable infrared power. (b) Short-circuit current density vs bandgap with AM1.5G on the left ordinate in mA cm^{-2} and the four 500-lux/low-light spectra on the right ordinate in μA cm^{-2}. The inset enlarges the LED, CFL, and low-light AM1.5G curves over 0–200 μA cm^{-2}; its horizontal axis is bandgap in eV. The full halogen curve remains on the outer right ordinate. (c) Open-circuit voltage vs bandgap. The indoor voltage penalty follows the logarithm of photon flux and is, therefore, modest compared with the reduction in current. (d) Ideal fill factor vs bandgap. The shaded range in every panel identifies 1.80–1.90 eV. All curves use step absorptivity, one emitting hemisphere, and a cell temperature of 300 K.

$E_g = 1.34$ eV, whereas the LED and CFL maxima shift to 1.94 and 1.95 eV and reach 54.06% and 55.66%, respectively. At $E_g = 1.85$ eV, the outdoor SQ limit is 26.10%, while the LED and CFL limits are 52.86% and 53.16% at 500 lux. A low-light AM1.5G spectrum retains a broad spectral mismatch and gives 23.58% at the same gap; the halogen spectrum is particularly unfavorable because most power is infrared, producing only 3.58% at the t-Se gap despite its high irradiance.

The current origin of these trends is explicit in Fig. 2(b). The one-sun J_{SC} decreases rapidly with increasing bandgap and is

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18.28 mA cm^{-2} at 1.85 eV; by contrast, LED and CFL currents are low in absolute magnitude but retain a high fraction of their incident photon flux near the optimum gap. Figure 2(c) shows that the higher bandgap yields a large radiative V_{OC}, 1.558 V outdoors at 300 K, while reduced indoor photon flux lowers voltage logarithmically rather than linearly. Figure 2(d) shows that the ideal fill factor (FF) remains above 90% for t-Se under all modeled spectra because the normalized open-circuit voltage is large. The experimentally observed FF deficit is, therefore, not imposed by the bandgap; it arises from nonradiative diode currents, incomplete collection, resistive loss, or voltage-dependent shunting.

C. Thermodynamic variables that shift the t-Se limit

The SQ limit depends on the stated optical and thermal conditions. Figure 3(a) separates the fixed-optical-edge temperature response from an illustrative optical-edge sensitivity. The slate and coral curves retain the 300 K step gap and experimental absorption spectrum, respectively; the blue and green curves additionally use $E_g(T) = E_g(300\,\mathrm{K}) + \gamma_g(T - 300\,\mathrm{K})$ and rigidly shift the absorption spectrum by the same energy. Here, $\gamma_g = -3.0 \times 10^{-4}$ eV K^{-1} is an assumed sensitivity coefficient, not a fit to Se measurements or a statistical uncertainty bound. Published Se photoluminescence measurements show sample-dependent temperature shifts; they do not calibrate the absorption spectrum used here.[14] An experimentally established temperature-dependent absorption spectrum for this same input film is not supplied. The two calculations, therefore, bracket selected modeling choices, not the full physical uncertainty. The small steps in the moving-gap curve reflect the 1-nm spectral grid; the numerical effect is quantified in the supplementary material.

External luminescence efficiency, η_{ext}, is a compact measure of the fraction of total recombination that produces photons escaping the device. Figure 3(b) shows the resulting efficiency penalty from $\Delta V_{OC} = (k_B T/q)\ln\eta_{ext}$. An eight-decade sweep reveals that indoor operation is not intrinsically tolerant of poor luminescence: its high ideal efficiency still requires suppression of nonradiative loss. Because V_{OC} is already reduced at low irradiance, the same η_{ext} can consume a larger fraction of the available voltage headroom indoors.

Figure 3(c) evaluates conditional angular-restriction sensitivity for AM1.5G, LED, CFL, halogen, and low-light AM1.5G spectra. The projected emission solid angle is $\Omega_{em} = \int_A \cos\theta\, d\Omega$, equal to π for one unrestricted hemisphere. With $C_\Omega = \pi/\Omega_{em}$, $J_{0,rad}(C_\Omega) = J_{0,rad}(1)/C_\Omega$. At fixed accepted generation, the ideal-diode voltage increase is $\Delta V_\Omega = (k_B T_c/q)\ln[(1 + C_\Omega J_{gen}/J_0)/(1 + J_{gen}/J_0)]$, approaching $(k_B T_c/q)\ln C_\Omega$ for $J_{gen} \gg J_0$. Reciprocity also restricts illumination acceptance. More generally, if accepted generation changes by a factor f_{acc}, the large-photocurrent voltage gain becomes $(k_B T_c/q)\ln(f_{acc} C_\Omega)$. Thus, a diffuse uniform source with $f_{acc} = 1/C_\Omega$ provides no such voltage gain.[7,8,24] The plotted indoor curves hold accepted generation fixed and are not realizable gains for an arbitrary room-light angular distribution. Likewise, AM1.5G contains diffuse light; the 4.6×10^4 marker refers only to the direct-sun geometric concentration benchmark.

Figure 3(d) maps the coupled dependence on the independently specified bandgap and temperature. The dashed trajectory overlays the same illustrative $E_g(T)$ sensitivity used in panel (a), while the surrounding map remains useful for materials whose gap differs from the 300 K reference. A cooler module helps throughout the t-Se gap range, but it does not move 1.85 eV to the one-sun optimum. Detailed formulations for emission geometry and angular restriction are given in Sec. S1 of the supplementary material; temperature-dependent optical sensitivity is detailed in Sec. S4 of the supplementary material; and refractive-index and doping sensitivity are given in Sec. S5 of the supplementary material.

D. Finite absorption, thickness, and light trapping

The step-function model conceals a central t-Se problem: the optical edge is not only uncertain by approximately 0.2 eV across reports, but is also broad on the energy scale that controls reciprocal emission. Figure 4(a) compares the representative absorption coefficient with rigid edge shifts of ± 0.05 eV, approximating the experimental spread. A red-shift increases J_{SC} but also increases the radiative saturation current; a blue-shift does the opposite. Thus, a lower "Tauc gap" does not necessarily improve efficiency unless the edge remains steep.

Figure 4(b) converts the absorption coefficient into two-pass absorptivity. At high photon energy, even a 50 nm film is strongly absorbing. Near 1.8 eV, however, the absorptivity continues to improve up to micrometer thicknesses. This reconciles the approximately 100 nm films that can show high blue external quantum efficiency with the approximately 1 μm thickness found by Hadar *et al.* to maximize current near the band edge.[11,12]

Figure 4(c) shows the resulting radiative-limit efficiency for single-pass, planar two-pass, and ideal ergodic-texturing optical models. The one-micrometer-thick planar limit is 26.71%; the ideal textured limit is 28.00%. The latter exceeds the step-absorber value quoted at a nominal 1.85 eV because the experimentally derived spectrum absorbs below that nominal edge and, therefore, behaves as a lower effective photovoltaic gap. This is not a violation of detailed balance; the global SQ envelope remains higher. Figure 4(d) separates the current and voltage consequences. Light trapping increases current at small thickness, but the increased reciprocal emissivity can slightly reduce voltage. The optimum, therefore, occurs when current saturation, recombination volume, and photon escape are treated together rather than by maximizing absorptance alone. Related detailed-balance analyses of plasmonic metamaterial absorbers and realistic perovskite/silicon tandem stacks likewise show that optical structures must be judged by their net device-level balance rather than by absorption enhancement alone; even an ideal intermediate mirror need not improve an already optimized stack.[25,26] The edge-shift, imposed-Urbach-energy, and digitization uncertainty tests are reported in Sec. S4 of the supplementary material.

E. Recombination quality, Auger loss, rear reflectivity, and photon escape

Figure 5(a) establishes the Se-specific dependence of efficiency on effective Shockley–Read–Hall lifetime. The lifetime is

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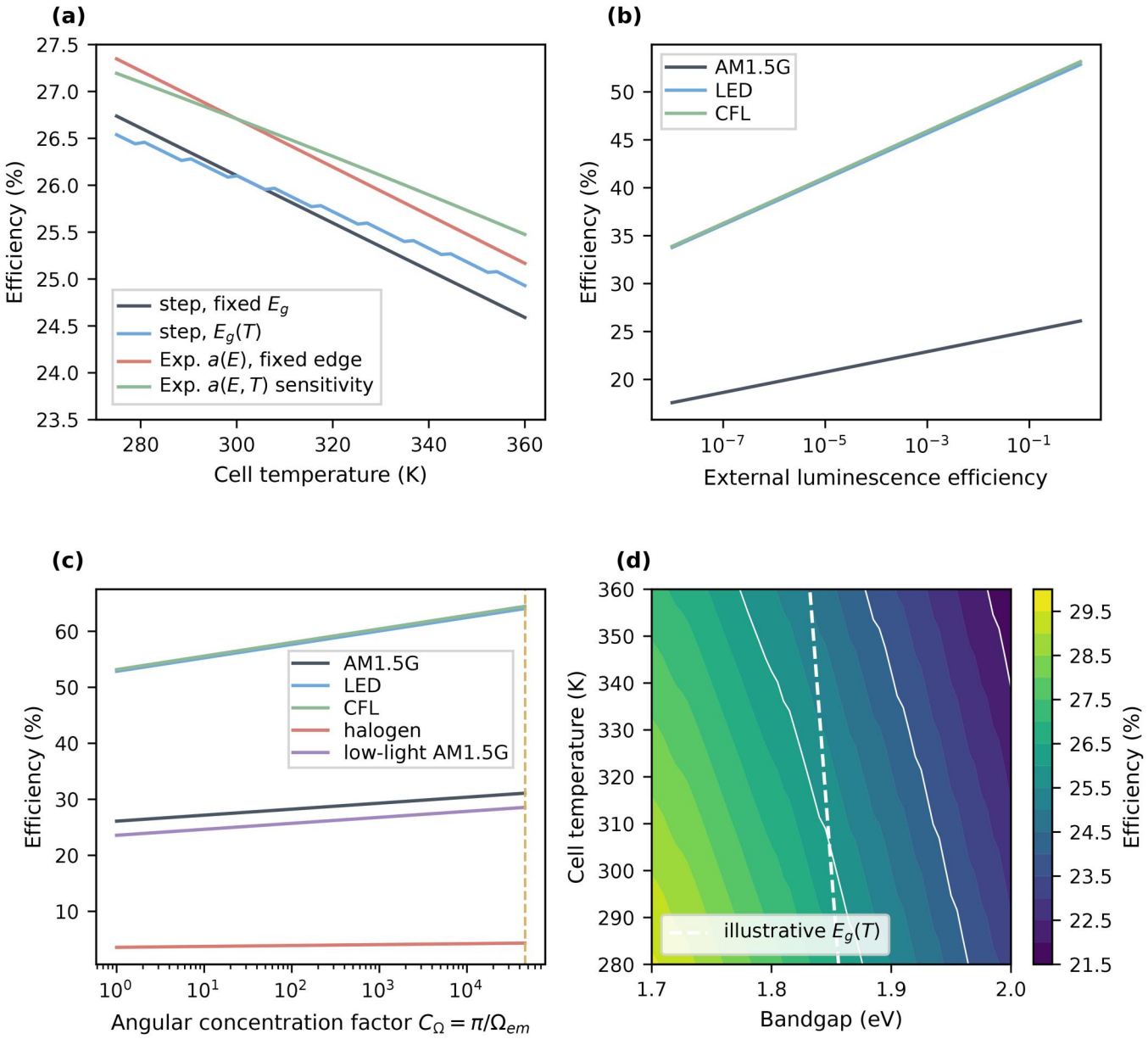


FIG. 3. Thermodynamic sensitivities of t-Se efficiency. (a) One-sun efficiency vs temperature: slate, fixed step gap; blue, temperature-shifted step gap; coral, fixed experimental absorption spectrum; and green, rigidly shifted experimental spectrum. All four curves are solid. "Exp." denotes experimental. The assumed linear shift is $\gamma_g = -3.0 \times 10^{-4}$ eV K^{-1} relative to 300 K; it is not a measured Se coefficient. (b) Efficiency vs external luminescence efficiency for AM1.5G, LED, and CFL spectra under an ideality-one loss model. (c) Efficiency vs $C_\Omega = \pi/\Omega_{em}$ for all five illumination spectra, holding accepted generation fixed. This conditional result requires compatible illumination acceptance and does not predict a free gain for diffuse room light. The vertical line marks the direct-sun concentration benchmark. (d) Efficiency vs independently specified gap and temperature; the dashed trajectory is the illustrative $E_g(T)$ path. No calibrated temperature-dependent absorption spectrum is available for the input film.

swept from picoseconds to milliseconds using the t-Se intrinsic carrier concentration $n_i = 1.09 \times 10^4$ cm^{-3} evaluated at 300 K from the integrated densities of states and the adopted gap. The primary analysis gives $N_C = 8.7 \times 10^{18}$ and $N_V = 1.64 \times 10^{20}$ cm^{-3}; the later review combines those values with a 1.85 eV photoluminescence gap to obtain the adopted n_i.[14,27] For the intrinsic baseline plotted here, efficiency approaches its radiative limit as the effective lifetime enters the tens-of-microseconds to submillisecond range.

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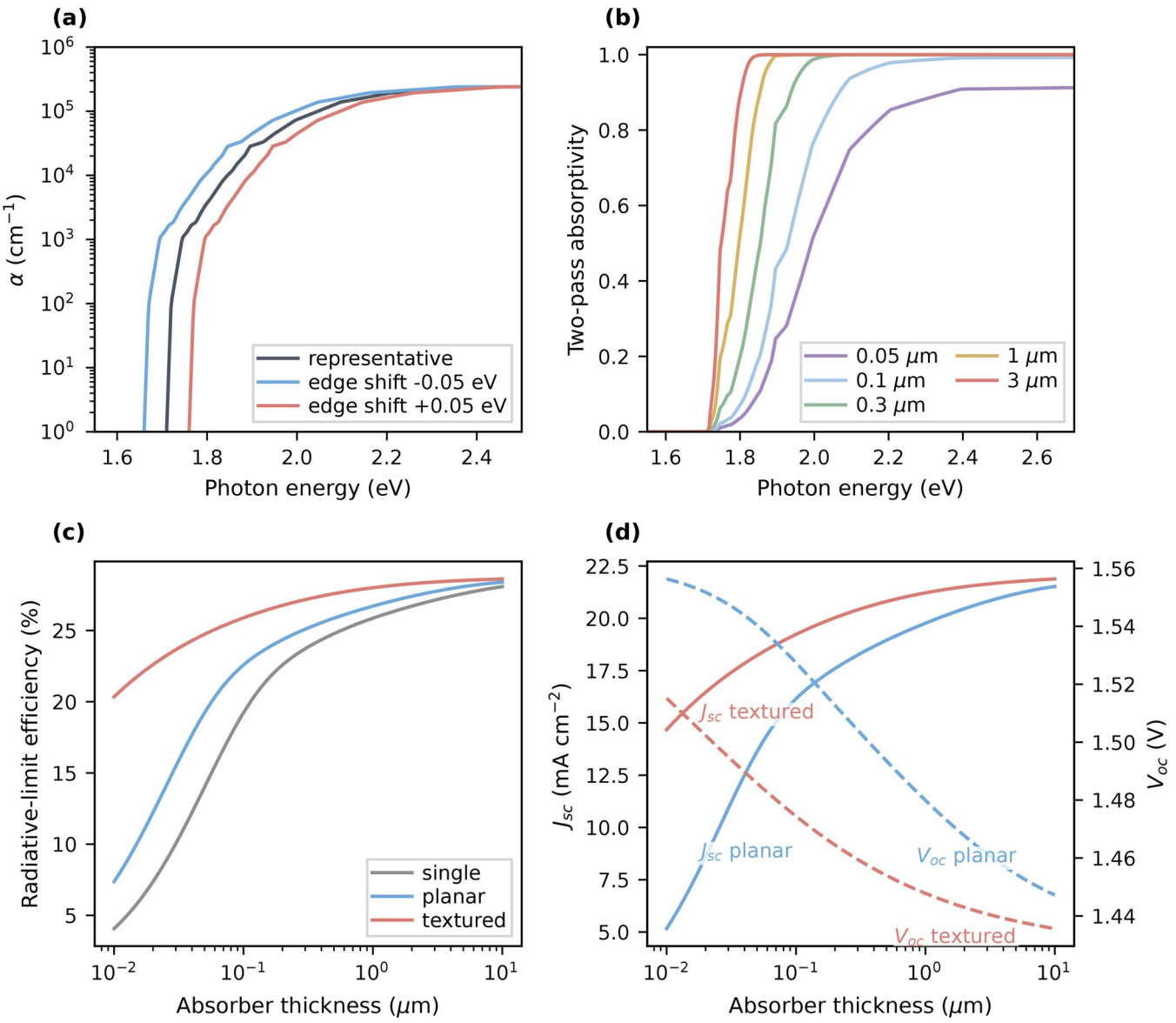


FIG. 4. Material-specific optical limits of t-Se. (a) Representative absorption coefficient and rigid edge shifts of ± 0.05 eV, spanning a realistic portion of the literature variation. (b) Two-pass absorptivity for t-Se thicknesses between 0.05 and 3 μ m. (c) One-sun radiative-limit efficiency vs thickness for single-pass, planar two-pass, and ideal textured light trapping. (d) Short-circuit current density (solid curves, left axis) and open-circuit voltage (dashed curves, right axis) vs thickness for planar and textured optics. Current gain and voltage reduction occur simultaneously because absorptivity and emissivity are reciprocal.

This is not a doping-independent lifetime requirement. The record study reports a 22.9-ns transient decay, but a measured decay is not automatically the same parameter as the effective recombination lifetime used here.[13] The review cautions that reported t-Se transients span picoseconds to microseconds and may reflect localization, trapped carriers, or spatially separated charge rather than one unique bulk lifetime; the sweep should, therefore, be read as a performance requirement, not as an assignment of a specific experiment.

Figure 5(b) sweeps the effective Auger coefficient. No consensus t-Se Auger coefficient is available, so the graph deliberately reports the threshold at which Auger recombination becomes relevant. In the radiative/SRH-free case, coefficients below roughly 10^{-27} cm^6 s^{-1} have little effect under one sun, while larger values

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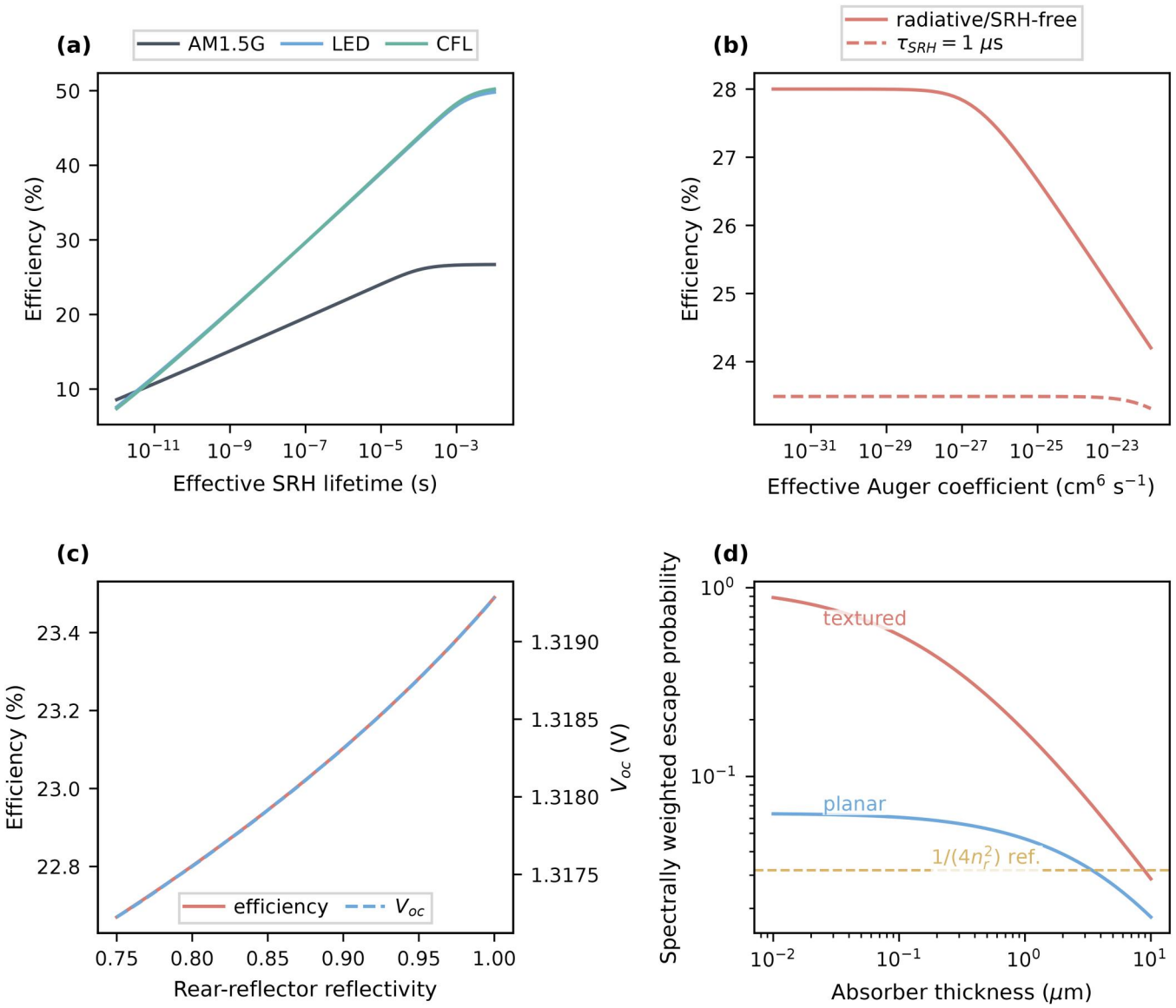


FIG. 5. Recombination quality and photon management. (a) Efficiency vs effective Shockley–Read–Hall (SRH) lifetime under one-sun, LED, and CFL illumination for a one-micrometer-thick planar t-Se absorber. (b) One-sun efficiency vs effective Auger coefficient for a radiative/SRH-free absorber and for an effective SRH lifetime of $1\,\mu$ s. (c) One-sun efficiency (left axis) and open-circuit voltage (right axis) vs rear-reflector reflectivity for a textured absorber with a $1\,\mu$ s lifetime. (d) Spectrally weighted escape probability vs absorber thickness for planar and textured optics. The dashed horizontal line is the geometric reference $1/(4n_r^2)$ at $n_r = 2.8$, not an asymptote of the plotted spectral ratio.

reduce efficiency. With an effective $1\,\mu$ s SRH lifetime, SRH loss dominates over a wider Auger range. This establishes an experimental priority: Auger metrology becomes decisive only after the defect-mediated lifetime has been improved sufficiently.

Rear reflectivity has a dual role. Figure 5(c) shows that increasing reflector quality raises light trapping and reduces parasitic luminescence absorption, improving both efficiency and V_{OC}. An optically reflective but electrically lossy metal cannot deliver this gain if it introduces contact recombination. Figure 5(d) calculates the spectrally weighted photon-escape probability vs thickness. P_{esc} is the equilibrium external-emission flux divided by the total internal radiative generation integrated through the absorber. It measures escape per radiative generation event and is not the geometric escape fraction at a single surface encounter. In the

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weak-absorption limit, the present planar two-pass model gives $P_{esc} \to 1/(2n_r^2)$, whereas ideal texturing gives $P_{esc} \to 1$. Both can decrease as reabsorption strengthens with thickness. The line $1/(4n_r^2)$ is a geometric reference, not the thick-film asymptote of this spectrally weighted quantity.[6,7] From the representative absorption coefficient, refractive index $n_r = 2.8$, and n_i, the van Roosbroeck–Shockley relation gives an internal radiative coefficient $B_{int} \approx 3.06 \times 10^{-11}$ cm^3 s^{-1} at 300 K. Its absolute value is model-derived and should be refined when polarization-resolved t-Se optical constants become available. Full SRH, Auger, doping, and internal/external luminescence derivations are in Sec. S6 of the supplementary material. The resistance and contact implementation used to convert junction losses into terminal current–voltage curves is introduced in Sec. S7 of the supplementary material.

F. Indoor operation: efficiency, output power, and spectral loss partition

Figure 6(a) maps LED efficiency from 1 to approximately 2000 lux. In the radiative limit, efficiency increases modestly as voltage grows logarithmically with illuminance. The finite-lifetime curves use the full SRH expression but retain an intrinsic baseline, $N_A = 0$. Their stronger intensity dependence is, therefore, conditional on that carrier-density regime, not a universal property of SRH-limited Se devices. At finite p-type doping and low injection, SRH recombination approaches an ideality-one dependence, and its ratio to radiative recombination can become nearly intensity-independent. The explicit doping/injection audit in Sec. S6 of the supplementary material demonstrates this distinction. Transfer of a measured 1000-lux efficiency to 200 lux still requires a current–voltage calculation that includes doping, recombination, collection, and leakage.

Figure 6(b) separates V_{OC} and FF for the 1 μ s case. Voltage rises approximately linearly with $\ln(L_v/L_{v,ref})$, the logarithm of the illuminance ratio to a fixed reference, while FF follows the normalized voltage. Series resistance is less restrictive at low current, but leakage through finite shunt resistance becomes disproportionately important. Figure 6(c) converts efficiency into maximum electrical power density and overlays approximate application ranges. At nominal 500 lux, the radiative t-Se LED output is of order 90 μ W cm^{-2}, while realistic outputs depend on nonradiative quality and active area. The shaded power-density ranges are indicative, area-dependent application guides, not specifications for particular wireless protocols. The finite-lifetime curves isolate SRH loss and do not equal the contact-limited advanced-target scenario introduced below.

Figure 6(d) partitions incident power for a 1.85 eV step absorber. LED and CFL illumination allocate only 1.30% and 4.22% to sub-bandgap photons, respectively; their dominant irreversibility is voltage/entropy loss plus modest thermalization. Halogen illumination allocates 94.0% below the gap, explaining its poor t-Se efficiency despite 500 lux. Low-light AM1.5G retains the same 56.3% sub-bandgap fraction as one sun because illuminance scaling changes intensity but not spectral shape. The photometric normalization described above is used throughout; intensity and area scaling are treated in Sec. S8 of the supplementary material.

G. From the certified record to practical outdoor and indoor limits

A practical-limit analysis should be checked against independent performance metrics of an existing device before constructing parameter sensitivities. Figure 7(a) compares radiative, advanced-target, and record-equivalent one-sun current–voltage curves. The certified record is $J_{SC} = 14.20$ mA cm^{-2}, $V_{OC} = 1.03$ V, FF=70.25%, and efficiency=10.3%.[13,14] A non-unique inverse calibration fits the three independent metrics J_{SC}, V_{OC}, and FF (whose product determines efficiency) using a reciprocal external-response factor $f_{ER} = 0.725$, an effective SRH lifetime of 5.55 ns, a series resistance of 8.40 Ω cm^2, a shunt resistance of 10^4 Ω cm^2, and a rear reflectivity of 0.8. The factor f_{ER} scales the modeled external absorptivity/emissivity and is not a measured carrier-collection efficiency. These are *equivalent lumped parameters*, not uniquely extracted material constants; other combinations of interface recombination, voltage-dependent collection, and contact barriers can reproduce the same summary metrics; the full measured current–voltage trace was not independently fitted.

Figure 7(b) compares five explicitly defined scenarios under one sun and 500-lux LED illumination. The radiative planar and textured cases isolate optical architecture. The record-equivalent case retains the calibrated loss parameters without claiming that the record outdoor device is optimized for indoor use. The advanced-target case assumes ideal texturing with $R_b = 0.95$, a 1 μ s lifetime, $R_s = 1$ Ω cm^2, $R_{sh} = 10^5$ Ω cm^2, and $f_{ER} = 0.90$; it reaches 20.5% outdoors and 26.5% under the LED. The practical-limit scenario assumes $R_b = 0.99$, $\tau = 100\,\mu$s, $R_s = 0.2$ Ω cm^2, $R_{sh} = 10^6$ Ω cm^2, and $f_{ER} = 0.98$; it reaches 26.9% and 42.2%, respectively. These numbers are parameter-conditioned limits at 300 K with $N_A = 0$, not forecasts or experimentally established joint material-property combinations. The microscopic origin of the dominant Se recombination remains unresolved, so no development timescale is assigned to either target. Calculations of intrinsic point-defect tolerance do not establish the recombination kinetics of processed device interfaces.[28]

Figure 7(c) maps outdoor efficiency against effective lifetime and series resistance for the advanced-target optical stack. At short lifetime, reducing resistance alone yields little benefit; once lifetime exceeds approximately the microsecond scale, contact resistance becomes the main barrier to efficiencies above 20%. Figure 7(d) maps indoor efficiency against lifetime and shunt resistance at a deliberately conservative series resistance. Leakage is decisive at low generation: high indoor efficiency requires both long lifetime and R_{sh} well above 10^5 Ω cm^2. The series-resistance, shunt-resistance, and external-response formulations are given in Sec. S7 of the supplementary material.

H. Sub-bandgap reflection, module temperature, and nonideal optical trade-offs

Figure 8(a) quantifies the source-dependent sub-bandgap power fraction. For t-Se at 1.85 eV, it is 56.3% under AM1.5G but only 1.30% under the LED and 4.22% under the CFL. Halogen illumination is the opposite extreme at 94.0%. These fractions concern incident energy, not necessarily absorbed heat: a

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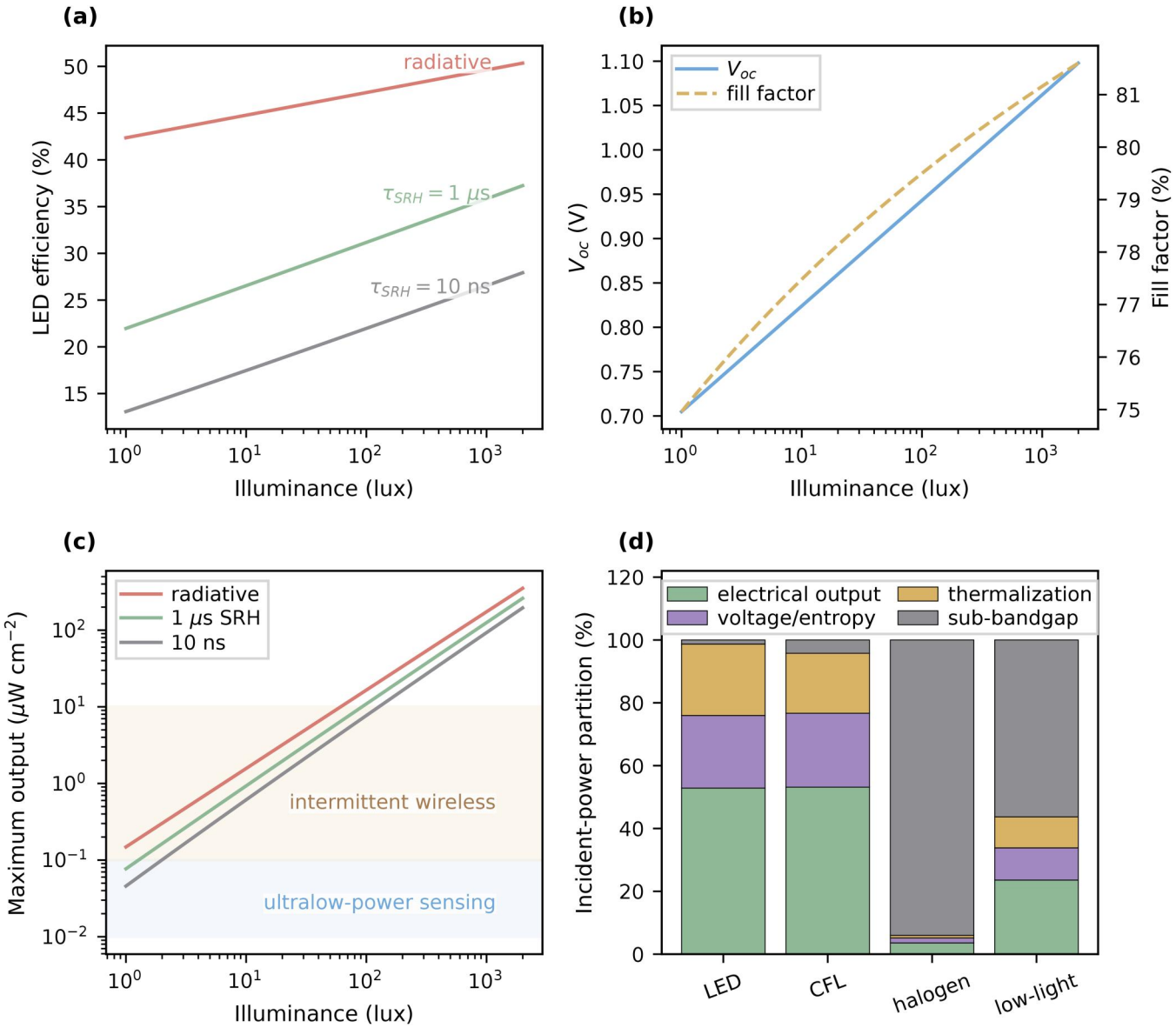


FIG. 6. Indoor efficiency and usable power. (a) LED power-conversion efficiency vs illuminance for a radiative absorber and for effective SRH lifetimes of 1 μ s and 10 ns. (b) Open-circuit voltage and fill factor vs illuminance for the 1 μ s case. (c) Maximum output power density vs illuminance for radiative and finite-lifetime devices, with indicative power-density ranges for ultralow-power sensing and intermittent wireless operation. All curves in (a)–(c) use a one-micrometer-thick, ideally textured intrinsic absorber with complete extraction and $f_{ER} = 1$, a perfect rear reflector, and no series or shunt loss. (d) Incident-power partition at $E_g = 1.85$ eV into electrical output, voltage/entropy loss, thermalization, and sub-bandgap power for LED, CFL, halogen, and low-light AM1.5G spectra.

transparent module can transmit sub-bandgap light, and a reflective rear contact can reject part of it without a dedicated front filter.

Figure 8(b) applies the Sandia empirical module-temperature relation to an upper-bound opaque-module case in which nominal sub-bandgap power would otherwise be absorbed. At 25 °C ambient temperature and 1 m s^{-1} wind, the open-rack cell temperature is 57.3 °C without an SBR and 39.1 °C with ideal rejection. A close-roof configuration changes from 74.5 to 46.6°. These large reductions are a consequence of the unusually high t-Se cutoff energy and should not be applied to a semitransparent stack without measuring its actual parasitic absorptance.

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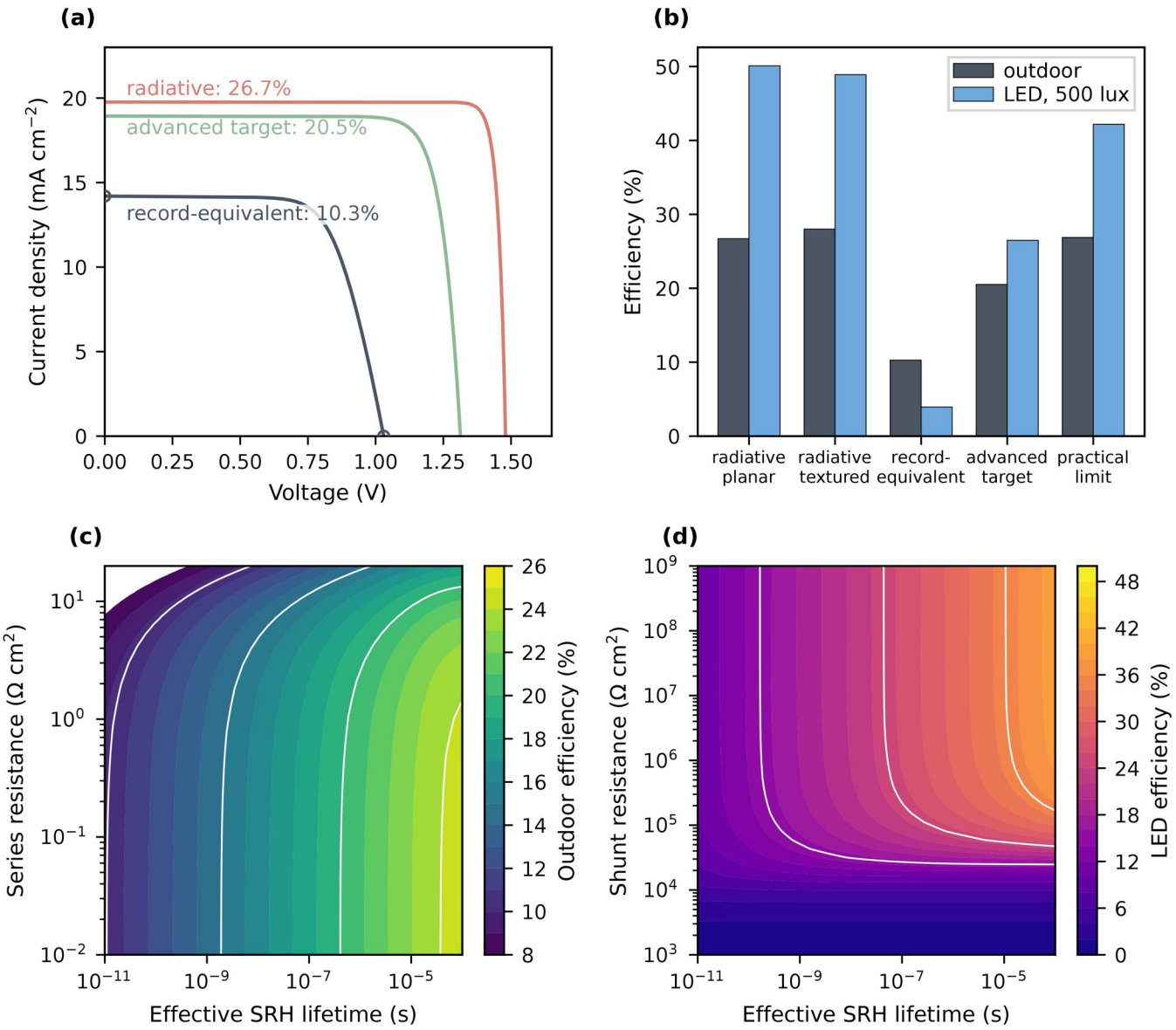


FIG. 7. Practical outdoor and indoor efficiency limits. (a) One-sun current–voltage curves for the radiative planar, advanced-target, and record-equivalent t-Se scenarios. Open symbols mark the certified record values used in the inverse calibration. (b) Outdoor and 500-lux LED efficiencies for radiative planar, radiative textured, record-equivalent, advanced-target, and practical-limit scenarios. Assumptions are stated in the text and tabulated in the supplementary material. (c) Outdoor efficiency map vs effective SRH lifetime and series resistance for a textured stack with $R_b = 0.95$, $f_{ER} = 0.90$, and $R_{sh} = 10^5\ \Omega\,\mathrm{cm}^2$. White contours include 10.3%, 15%, 20%, and 24%. (d) LED efficiency map vs effective SRH lifetime and shunt resistance for the same optical stack. White contours include the reported 15.1% indoor t-Se efficiency and higher target levels.

The isolated electrical response to cell temperature is quantified in Fig. 3(a). Figure 8(c) combines nonideal optics and thermal feedback: the horizontal coordinate is above-bandgap transmittance into the cell, while the vertical coordinate is sub-bandgap reflectance that reduces thermal load. High SBR reflectance is beneficial only if it is achieved without sacrificing above-bandgap throughput. For t-Se, a few percent visible loss can offset a substantial fraction of the electrical cooling gain, so an eventual deployment study should optimize electrical energy yield rather than infrared reflectance alone. The present map is an instantaneous, empirical thermal sensitivity, not a weather-resolved annual-yield calculation. The optical-temperature coupling,

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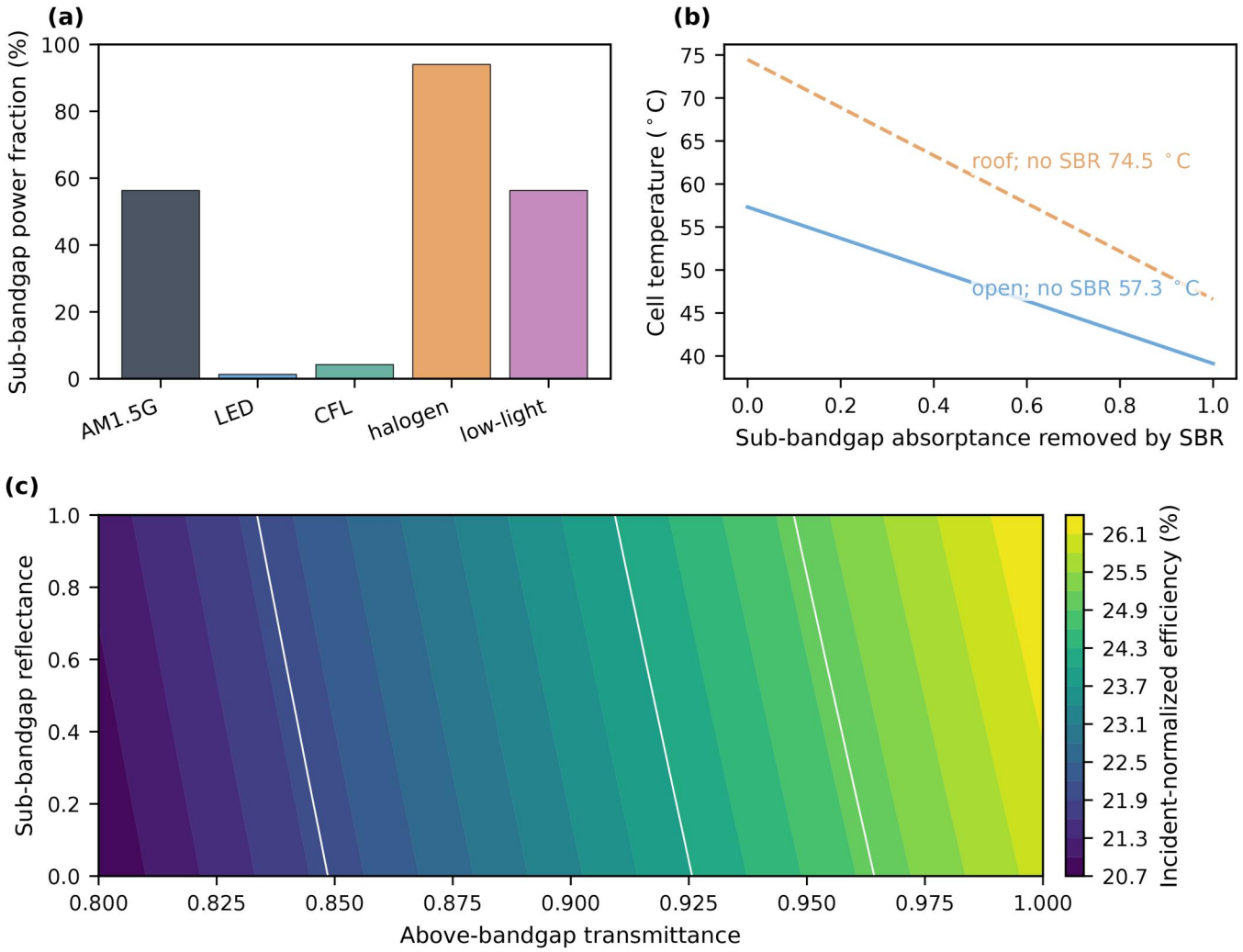


FIG. 8. Sub-bandgap-reflector and temperature limits. (a) Fraction of incident power below a 1.85 eV bandgap for AM1.5G, LED, CFL, halogen, and low-light AM1.5G spectra. (b) Sandia-model cell temperature vs the fraction of nominal sub-bandgap absorptance removed by a reflector for open-rack and close-roof mounting at 25 °C ambient temperature and 1 m s^{-1} wind. (c) Incident-normalized efficiency as a joint function of a scalar useful-photon throughput and sub-bandgap reflectance. This separate radiative planar sensitivity uses fixed 300-K optical-edge data, complete extraction apart from that throughput, and no explicit SRH or resistive loss. The map exposes the trade-off between optical current loss and thermal voltage gain. The separate efficiency-temperature dependence is given in Fig. 3(a).

mounting parameters, and sensitivity to ambient conditions are detailed in Sec. S9 of the supplementary material.

I. Temperature-accelerated degradation and conditional performance targets

Thermal management affects energy yield twice: it changes instantaneous efficiency and can change degradation rate. Figure 9(a) plots Arrhenius rate acceleration relative to 25 °C for activation energies of 0.3, 0.6, and 0.9 eV. As a validated t-Se module activation energy is not yet available, these are sensitivity parameters rather than fitted kinetics. Figure 9(b) converts the acceleration factors into illustrative t_{80} lifetimes for baseline 25 °C degradation rates of 0.5, 1, and 3% yr^{-1}. Indoor LED operation is represented by ambient temperature because irradiance-driven self-heating is negligible; outdoor mounting and SBR cases use the temperatures from Fig. 8(b).

Figure 9(c) integrates normalized output over 30 years for an illustrative 1% yr^{-1} rate at 25 °C and activation energy 0.6 eV. An ideal SBR increases lifetime yield by lowering temperature, but the absolute result should not be interpreted as a t-Se warranty prediction until dominant degradation mechanisms and activation energies are measured. Figure 9(d) summarizes the performance ladder. Outdoors, the major steps involve the recovery of the

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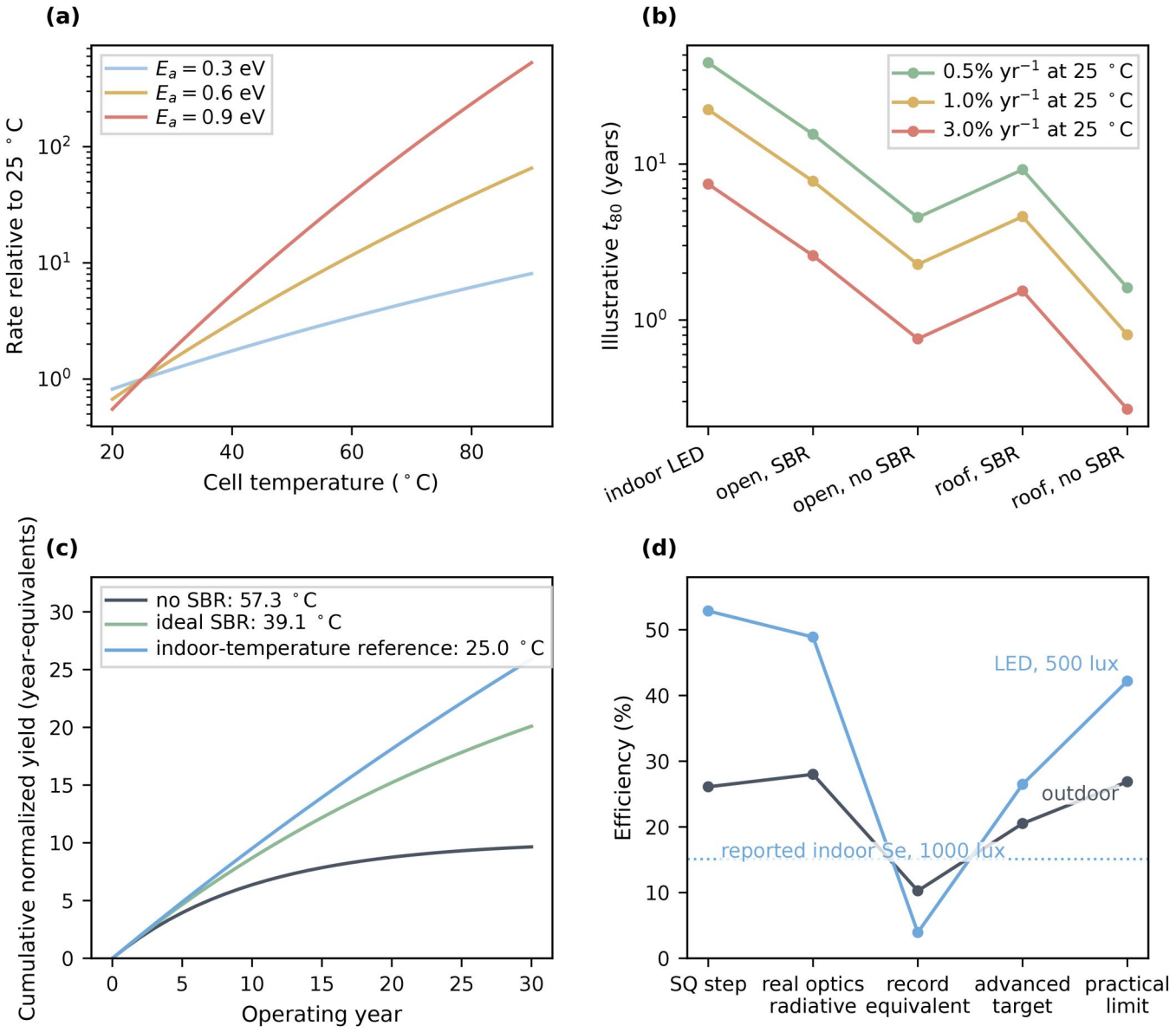


FIG. 9. Illustrative degradation sensitivity and conditional t-Se performance targets. (a) Arrhenius degradation-rate ratio relative to 25 °C for activation energies of 0.3, 0.6, and 0.9 eV. (b) Illustrative time to 80% remaining performance, t_{80}, for indoor-temperature, open-rack, and close-roof configurations with and without ideal sub-bandgap reflection. Curves correspond to three assumed baseline rates at 25 °C. (c) Thirty-year cumulative normalized yield for an illustrative 1% yr^{-1} baseline rate and 0.6 eV activation energy. (d) Outdoor and LED efficiency ladder from the step SQ value through the radiative limit based on experimental absorptivity, record-equivalent, advanced-target, and practical-limit scenarios. The dotted line marks the reported 15.1% indoor t-Se efficiency measured at 1000 lux.

absorption-limited current, nonradiative voltage, and FF. Indoors, the high SQ limit is countered by sensitivity to the injection-dependent SRH rate and to leakage at low irradiance. The reported 15.1% indoor result at 1000 lux is shown only as an experimental reference, not as a point on the 500-lux scenario curve. The full degradation equations and uncertainty maps are provided in Sec. S10 of the supplementary material.

III. DISCUSSION

A. Why t-Se is simultaneously an imperfect outdoor absorber and an unusually well-matched indoor absorber

The thermodynamic distinction between outdoor and indoor t-Se is primarily spectral, not chemical. Under AM1.5G, a 1.85 eV

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gap rejects more than half of incident power before any device loss occurs and thermalizes a portion of the accepted high-energy power. The result is a 26.1% step-absorber limit, substantially below the global single-junction maximum. Under a white LED or CFL, the same gap admits most photons while avoiding much of the low-energy spectral tail, so the limit rises above 50%. This supports indoor and tandem-top-cell deployment rather than replacing crystalline silicon in unconcentrated single-junction modules.

The absence of strong near-infrared content indoors has two separate consequences that should not be conflated. First, it improves conversion efficiency because less incident energy is counted in the denominator without being absorbed. Second, it suppresses irradiance-driven heating. At 500 lux, LED and CFL power densities are only approximately 1.4–1.8 W m^{-2}, far below one sun. Even complete absorption would produce a temperature rise much smaller than normal room-to-room ambient variation. The modeled indoor thermal advantage arises primarily from low irradiance and modest temperature rather than from an SBR; this calculation does not establish chemical or operational stability under a particular indoor atmosphere. Halogen lighting is an exception spectrally but still produces only a small module temperature rise at 500 lux.

B. The photovoltaic edge, not the quoted optical gap, is the controlling material descriptor

A single t-Se bandgap value is insufficient for a quantitative limit.[29] The review compiles Tauc gaps between 1.8 and 2.0 eV, a photoluminescence peak near 1.85 eV, and an external-quantum-efficiency onset shifted to lower energy.[14] In detailed balance, every subgap absorptivity increment contributes to both photocurrent and emission. A broad edge can raise J_{SC} while lowering V_{OC}, and the voltage loss is amplified when nonradiative recombination is present. Therefore, experiments should report absorptance, external quantum efficiency, electroluminescence, and photoluminescence on the same device and use reciprocity to reconstruct the radiative saturation current.[4,30]

The radiative calculation also clarifies why "strong absorption above 10^5 cm^{-1}" does not guarantee an ultrathin optimum. High-energy absorption saturates rapidly, but the current near the wide t-Se edge requires hundreds of nanometers to micrometers unless light trapping is effective. Conversely, increasing thickness adds recombination volume and can reduce escape probability. The correct thickness is, therefore, quality-dependent: a poor-lifetime absorber should remain thin and optically trapped; a long-lifetime absorber can exploit thicker near-edge absorption without a severe nonradiative penalty.

C. A hierarchy of experimental priorities

The calculations establish a sequence rather than a single target. First, the device-relevant free-carrier lifetime must be separated from localization and trapped-charge decay by matched-injection optical-pump terahertz-probe, photo-Hall, transient photoluminescence, and quasi-steady-state photoluminescence measurements.[14,31] Second, quasi-Fermi-level splitting measured in an absorber/transport-layer stack should be compared with terminal V_{OC} to distinguish bulk/interface recombination from contact selectivity. Third, voltage-dependent external quantum efficiency and light-intensity-dependent shunt resistance should be measured because the 2019 study observed an illumination-induced shunt reduction that directly suppresses FF.[12,22] Fourth, rear reflectivity and luminescence extraction must be measured spectrally rather than assumed from metal identity.

A meaningful t-Se practical limit requires simultaneous improvement of these coordinates. The advanced-target 20.5% outdoor scenario is not achieved by any single parameter; it combines microsecond lifetime, $f_{ER} = 0.90$, high rear reflectivity, and low series resistance. The 26.9% practical limit approaches the radiative limit based on the experimental absorptivity and, therefore, requires both long lifetime and near-lossless contacts. Indoors, the same hierarchy shifts toward shunt resistance and nonradiative recombination because current is low. An indoor device can tolerate higher absolute series resistance, but it cannot tolerate leakage comparable to its sub-milliampere-per-square-centimeter photocurrent.

D. Interpretation boundaries

Several assumptions define the scope of the numerical values. The indoor spectra are measured or reconstructed reference spectra normalized to 500 lux; other correlated color temperatures and luminaires will produce different powers and optima. The representative t-Se absorption coefficient is a digitized experimental curve, not a universal dielectric function. The refractive index is held at 2.8 for photon-escape estimates. Carrier concentrations and quasi-Fermi-level splitting are spatially uniform in the recombination model, and complete extraction is assumed unless the empirical external-response factor f_{ER} is specified. The model, therefore, omits position-dependent generation, electric field, mobility, diffusion, depletion width, and contact boundary conditions. The resulting radiative and recombination maps are zero-dimensional optical–recombination upper limits; low and anisotropic Se mobility can lower current and fill factor in a real junction. The scalar f_{ER} is a lumped fit coordinate, not a physical collection efficiency. The Auger coefficient, activation energy, and long-term degradation rate are parameter sweeps because these quantities are not calibrated for the modeled device. The Sandia thermal model is empirical, and the ideal-SBR temperature reduction is an upper bound for an opaque module that would otherwise absorb sub-bandgap power. These assumptions and numerical verification against published benchmark values are summarized in Sec. S11 of the supplementary material.

IV. METHODS

A. Spectral and detailed-balance model

In the spectral equations, λ denotes the vacuum wavelength, I_λ is the spectral irradiance in W m^{-2} nm^{-1}, and T_c is the cell temperature. The incident spectral photon-flux density is

$$\Phi_\lambda^{in} = \frac{I_\lambda \lambda}{hc}, \tag{1}$$

where h is Planck's constant and c is the speed of light; Φ_λ^{in} is expressed in photons m^{-2} s^{-1} nm^{-1}. The short-circuit current

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density for reciprocal external absorptivity $a(\lambda)$, under the complete-extraction approximation unless a separately defined external-response factor is stated, is

$$J_{SC} = q \int a(\lambda)\Phi_\lambda^{in}\, d\lambda, \tag{2}$$

where q is the elementary charge. The external radiative saturation current for emission into projected solid angle Ω_{em} is

$$J_{0,rad} = q\Omega_{em} \int a(\lambda)\Phi_\lambda^{bb}(T_c)\, d\lambda, \tag{3}$$

where $\Phi_\lambda^{bb}(T_c)$ is the equilibrium blackbody photon radiance per unit wavelength of a cell at temperature T_c. Here, $\Omega_{em} = \int_A \cos\theta\, d\Omega$ is the projected angular measure. One emitting front hemisphere corresponds to $\Omega_{em} = \pi$. Detailed balance and photovoltaic reciprocity then give the ideal-diode response,

$$J(V) = J_{SC} - J_{0,rad}\left[\exp\left(\frac{qV}{k_B T_c}\right) - 1\right], \tag{4}$$

where V is the terminal voltage and k_B is Boltzmann's constant.[1,4] Maximum power is obtained numerically without an empirical FF approximation. The full derivation and emission geometry are provided in the supplementary material.

B. Experimentally derived optical model

The representative absorption coefficient was digitized from the red experimental curve in Fig. 5(a) of the 2026 t-Se review, corresponding to Ref. 42 therein.[14,23] Interpolation was performed in $\ln\alpha$ vs photon energy. The three sub-onset anchors at 1.680, 1.700, and 1.720 eV are numerical tail assumptions, not measured points; their origin and limitations are stated in Sec. S4 of the supplementary material. For thickness d, the single-pass and planar two-pass absorptivities are

$$a_{single} = 1 - e^{-\alpha d}, \tag{5}$$

$$a_{planar} = (1 - e^{-\alpha d}) + R_b e^{-\alpha d}(1 - e^{-\alpha d}), \tag{6}$$

and the ideal ergodic-texturing approximation is

$$a_{textured} = \frac{4n_r^2 \alpha d}{4n_r^2 \alpha d + 1 + n_r^2(1 - R_b)}. \tag{7}$$

Here, α is the band-to-band absorption coefficient, d is the absorber thickness, R_b is the rear reflectivity, and n_r is the absorber refractive index. The planar expression follows Beer–Lambert propagation, and the textured expression is the ergodic light-trapping approximation.[6,32] The formulas are used reciprocally: the same $a(\lambda)$ determines generation and external emission. The ergodic expression is a weak-absorption, isotropic-ray approximation, most reliable for high rear reflectivity; it is not an exact optical solution for an arbitrary nanoscale texture.

C. Nonradiative and parasitic electrical currents

Shockley–Read–Hall (SRH) recombination is evaluated at arbitrary injection using

$$U_{SRH} = \frac{n_e p_h - n_i^2}{\tau_p(n_e + n_1) + \tau_n(p_h + p_1)}, \tag{8}$$

$$J_{SRH} = q\int_0^d U_{SRH}(x, V_{int})\, dx = qdU_{SRH}(V_{int}) \quad \text{for} \quad \text{uniform } n_e, p_h. \tag{9}$$

Here, n_e and p_h are electron and hole concentrations, n_i is the intrinsic carrier concentration, n_1 and p_1 specify trap energy, and τ_n and τ_p are electron and hole capture times.[33,34] The equality $J_{SRH} = qdU_{SRH}$ used in the numerical maps is, therefore, an explicit spatially uniform-carrier approximation. Equivalently, transport is assumed sufficiently rapid that a single quasi-Fermi-level splitting characterizes the absorber. This can fail in low-mobility Se cells with nonuniform optical generation, depletion fields, diffusion lengths, or contact boundary conditions. The baseline uses a midgap trap and $\tau_n = \tau_p = \tau_{eff}/2$, preserving the effective lifetime coordinate without imposing high injection in the rate expression. The baseline nonetheless sets $N_A = 0$; finite-doping indoor cases are separately evaluated in the supplementary material. Under the same uniform-carrier approximation, the symmetric effective Auger current is

$$J_{Auger} = qd\frac{C_A}{2}(n_e p_h - n_i^2)(n_e + p_h), \tag{10}$$

where C_A is the sum of equal electron- and hole-mediated coefficients. This expression vanishes at equilibrium and approaches $qdC_A n_i^3 \exp(3qV_{int}/2k_B T_c)$ in intrinsic high injection.[6,32] Carrier densities follow charge neutrality and $n_e p_h = n_i^2 \exp(qV_{int}/k_B T_c)$; $n_i = 1.09 \times 10^4\ \text{cm}^{-3}$ is used for the 300-K recombination maps. The model does not solve drift–diffusion transport, and complete extraction is the upper-limit default. Series resistance is introduced parametrically through $V_{term} = V_{int} - JR_s$ and shunt current through V_{int}/R_{sh}. An ideality-one external-luminescence penalty and parasitic mirror absorption are included as additional currents proportional to the radiative emission current. Detailed expressions are provided in the supplementary material.

D. Indoor normalization

Illuminance is defined by

$$L_v = K_m \int I_\lambda V_{phot}(\lambda)\, d\lambda, \tag{11}$$

where $K_m = 683\ \text{lm W}^{-1}$ is the maximum photopic luminous efficacy and $V_{phot}(\lambda)$ is the dimensionless International Commission on Illumination (CIE) photopic luminous-efficiency function.[35] The LED, CFL, halogen, and low-light AM1.5G files retain the nominal 500-lux photometric normalization supplied with the open dataset of Nitta *et al.*[18] Their powers were integrated directly,

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but no independent recalibration against a newly measured lamp or a separately tabulated CIE response is claimed. Other illuminances are obtained by multiplying the same source shape by $L_v/(500\,\text{lux})$. Their integrated irradiances differ by more than an order of magnitude, so efficiency comparisons always use electrical output divided by the full source-specific incident power.

E. Module temperature, SBR, and degradation

The module temperature follows the Sandia relation

$$T_m = T_a + I_{\text{eff}}\exp(a + bv_w), \tag{12}$$

$$T_c = T_m + \frac{I_{\text{eff}}}{I_{\text{ref}}}\Delta T. \tag{13}$$

Here, T_a, T_m, and T_c are ambient, module, and cell temperatures in degrees Celsius; I_{eff} is the effective thermal irradiance in W m^{-2}; v_w is the wind speed in m s^{-1}; a is the fitted intercept for the stated irradiance and temperature unit convention; b has units s m^{-1} so that bv_w is dimensionless; and ΔT is the cell–module offset at the reference irradiance $I_{\text{ref}} = 1000$ W m^{-2}. The open-rack parameters are $a = -3.47$, $b = -0.0594$ s m^{-1}, and $\Delta T = 3$ K; the close-roof parameters are $a = -2.98$, $b = -0.0471$ s m^{-1}, and $\Delta T = 1$ K.[20,21] The effective thermal irradiance is reduced by the product of sub-bandgap incident fraction and the fraction of otherwise absorbed sub-bandgap power removed by the SBR. Degradation-rate acceleration relative to T_{ref} is

$$\frac{k(T_c)}{k(T_{\text{ref}})} = \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_{\text{ref}}} - \frac{1}{T_c}\right)\right]. \tag{14}$$

In this equation, $k(T)$ is the degradation rate, E_a is the assumed activation energy, and both T_c and T_{ref} are absolute temperatures in kelvin. The reference is 298.15 K not the 300-K electrical benchmark. This kinetic treatment is illustrative because no mechanism-specific rate or activation energy is calibrated for the modeled Se module.[20]

V. CONCLUSIONS

A spectrum-resolved detailed-balance analysis shows that the wide bandgap of t-Se creates two distinct photovoltaic opportunities. Under one-sun AM1.5G illumination, the step-absorber limit at 1.85 eV is 26.10%, and more than half of the incident power is sub-bandgap. Under 500-lux LED and CFL illumination, the same bandgap is close to the spectrum-specific optimum and yields ideal limits of 52.86% and 53.16%, respectively. The high indoor efficiencies are, therefore, a consequence of spectral matching and a reduced unusable near-infrared denominator, not of a different conversion mechanism.

Replacing the ideal step with a representative experimental absorption coefficient changes both generation and reciprocal emission. The resulting planar and textured radiative limits are 26.71% and 28.00%, and the broad photovoltaic edge can increase current while reducing voltage. A record-equivalent inverse calibration further demonstrates that the certified 10.3% device cannot be represented by optical loss alone: the chosen calibration requires an external-response factor together with nonradiative and resistive loss coordinates; it does not uniquely prove their microscopic allocation. Explicitly conditioned advanced-target and practical-limit parameter sets reach 20.5%–26.9% outdoors and 26.5%–42.2% under 500-lux LED illumination, identifying lifetime, contact selectivity, leakage suppression, and rear-stack optics as coupled rather than independent design coordinates.

For opaque outdoor modules, ideal sub-bandgap rejection can strongly reduce the modeled thermal load because 56.3% of AM1.5G power lies below the Se gap. The corresponding fractions under the modeled LED and CFL spectra are only 1.30% and 4.22%, so reflector-enabled cooling is principally an outdoor strategy. Quantitative deployment predictions will require measured full-stack absorptance, temperature-dependent band-edge data, injection-resolved lifetime and mobility, field- and thickness-dependent carrier extraction, luminescence extraction, and mechanism-specific degradation kinetics. The equations, source spectra, digitized optical data, numerical outputs, and figure-generation workflow supplied with this work provide a reproducible basis for updating the limits as those measurements become available.

SUPPLEMENTARY MATERIAL

See the supplementary material for the complete detailed-balance derivation, spectral and photometric normalization, numerical-convergence tests, absorption-edge uncertainty, photon-escape and doping analyses, parasitic electrical-loss maps, practical-scenario definitions, thermal and degradation sensitivity analyses, verification against the published Se Shockley–Queisser benchmark, and recommended experiments that map directly onto the model parameters. The accompanying code-and-data archive contains the source spectra, digitized t-Se absorption data, processed numerical outputs, and reproducible Python and LaTeX workflows.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Seungwoo Lee: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Methodology (lead); Project administration (lead); Software (lead); Supervision (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

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DATA AVAILABILITY

The data that support the findings of this study are available within the article, its supplementary material, and the accompanying code-and-data archive. The archive contains the source spectra, digitized t-Se absorption data, processed numerical outputs, and the Python and LaTeX workflows used to reproduce the calculations, tables, and figures.

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